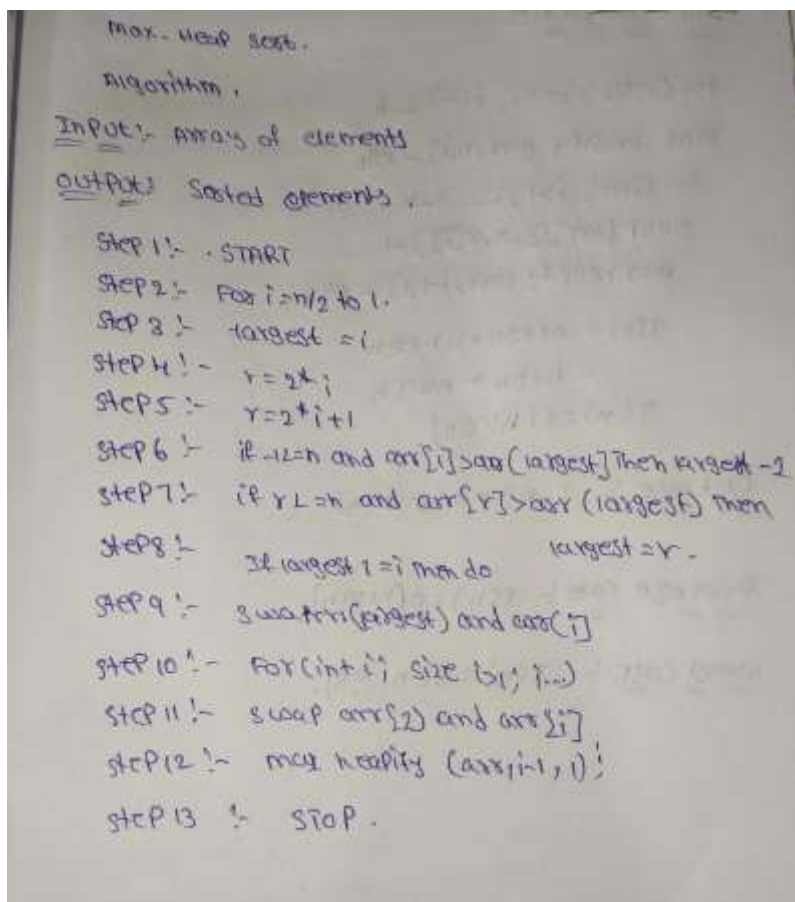


practical: 3	Implementation of max-heap sort algorithm
19/08/2026	

AIM: To implement the max-heap sort algorithm.

Algorithm for max-heap sort:



CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
void heapify(int arr[], int n, int i)
```

```
{
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
    if (left < n && arr[left] > arr[largest])
        largest = left;
    if (right < n && arr[right] > arr[largest])
        largest = right;
    if (largest != i)
    {
        swap(arr[i], arr[largest]);
        heapify(arr, n, largest);
    }
}

void heapSort(int arr[], int n)
{
    for (int i = n / 2 - 1; i >= 0; i--)
        heapify(arr, n, i);
    for (int i = n - 1; i > 0; i--)
    {
        swap(arr[0], arr[i]);
```

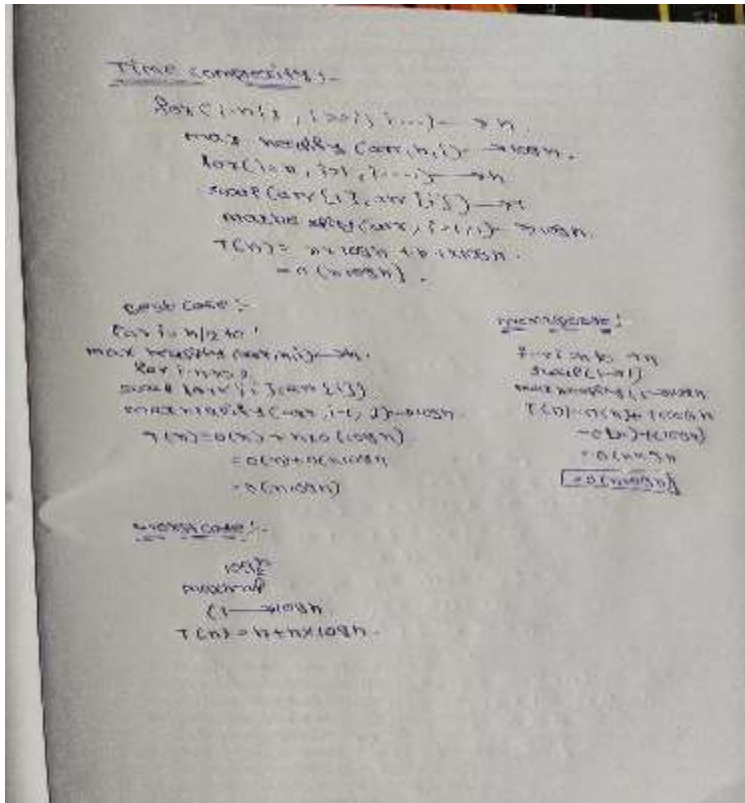
```
        heapify(arr, i, 0);
    }
}

int main()
{
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements: ";
    for (int i = 0; i < n; i++)
        cin >> arr[i];
    heapSort(arr, n);
    cout << "Sorted array: ";
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    return 0;
}
```

OUTPUT:

```
C:\Users\iswar\OneDrive\Desl  X  +  v
Enter number of elements: 7
Enter elements: 23 24 25 26 27 28 29
Sorted array: 23 24 25 26 27 28 29
```

TIME COMPLEXITY:



CONCLUSION:

Thus, Heap Sort using the Heapify algorithm was successfully implemented in C++. The array was converted into a Max Heap and the elements were repeatedly extracted to obtain the sorted array.